# REBot: Reflexive Evasion Robot for Instantaneous Dynamic Obstacle Avoidance

Zihao Xu, Ce Hao, Chunzheng Wang, Kuankuan Sima, Fan Shi, *Member, IEEE*, and Jin Song Dong, *Member, IEEE*

***Abstract*—Dynamic obstacle avoidance (DOA) is critical for quadrupedal robots operating in environments with moving obstacles or humans. Existing approaches typically rely on navigation-based trajectory replanning, which assumes sufficient reaction time and leading to fails when obstacles approach rapidly. In such scenarios, quadrupedal robots require reflexive evasion capabilities to perform instantaneous, low-latency maneuvers. This letter introduces Reflexive Evasion Robot (REBot), a control framework that enables quadrupedal robots to achieve real-time reflexive obstacle avoidance. REBot integrates an avoidance policy and a recovery policy within a finite-state machine. With carefully designed learning curricula and by incorporating regularization and adaptive rewards, REBot achieves robust evasion and rapid stabilization in instantaneous DOA tasks. We validate REBot through extensive simulations and real-world experiments, demonstrating notable improvements in avoidance success rates, energy efficiency, and robustness to fast-moving obstacles.**

***Index Terms*—Reinforcement learning, legged robots, collision avoidance.**

## I. Introduction

ENSURING the safety of a robot is essential during task execution.For instance, a mobile robot must not only perform its primary tasks but also perceive surrounding obstacles and take appropriate evasive actions [1], [2], [3]. When encountering slow-moving obstacles (reaction time $> 2$ s), the robot typically has sufficient time to stop its current actions and replan a new trajectory using a decision-making model to avoid collisions [4], [5]. This type of behavior is commonly referred to as dynamic obstacle avoidance (DOA) via navigation-based trajectory replanning [6]. For legged robots, such as quadrupedal robots, DOA involves both high-level navigation decision-making and low-level locomotion control [7]. For example, in the Agile but Safe (ABS) framework [8], a quadruped encountering a quasi-static obstacle during high-speed locomotion can reduce its speed and replan a new trajectory to safely bypass the obstacle.

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However, when obstacles approach at high speeds, the robot is left with limited reaction time ($< 1.5$ s), necessitating immediate evasive maneuvers [9]. Due to limitations in mechanical structure and motor power, the robot often fails to generate sufficient velocity within the available time to accurately track a replanned navigation trajectory [10]. To achieve instantaneous DOA, we draw inspiration from the spinal reflex systems. Unlike decision-making processes governed by the brain, spinal reflexes enable rapid, localized decisions through neural circuits in the spinal cord, allowing animals to execute unconventional evasive actions instantaneously. For example, an antelope might execute a sudden backward leap to evade an ambush from an underwater crocodile while drinking, relying entirely on reflexive evasion.

We propose the Reflexive Evasion Robot (REBot) for instantaneous dynamic obstacle avoidance. Using the Unitree Go2 as an example platform [11], [12], REBot demonstrates real-time evasion of high-speed obstacles with $< 1.5$ s reaction time. REBot is structured as a finite-state machine with three behavioral stages. In the normal stage, the robot performs its primary functional tasks. When an approaching obstacle is detected, REBot transitions to the avoidance stage, executing reflexive evasion maneuvers. During evasion, reinforcement learning (RL) policy [13], [14], [15] trained by PPO enables rapid avoidance while preserving safety, balance, and energy efficiency. After an evasive maneuver, the robot may become unstable. REBot then enters the recovery stage, during which a policy stabilizes the robot and restores normal function.

We trained the REBot system for quadrupedal robots in Isaac Gym simulator [16], evaluated its performance, and deployed it on a real robot for demonstration. REBot achieved the highest avoidance and recovery success rates in both static and dynamic obstacle scenarios, while reducing maximum joint power and avoidance distance. We observed that the robot's reflexive evasion performance varied with obstacle direction and speed; it performed best when avoiding frontal obstacles due to its structural advantages in backward maneuvers. Ablation studies confirmed that the recovery policy, curriculum learning, and adaptive reward design significantly improved avoidance success rates. Finally, real-world experiments (Fig. 1) validated REBot's capability for real-time, instantaneous dynamic obstacle avoidance, offering insights into robot safety design.

In summary, the contributions of this letter are as follows:

- We formally identify and formulate the reflexive evasion problem for dynamic obstacle avoidance under constrained reaction time in quadrupedal robots.

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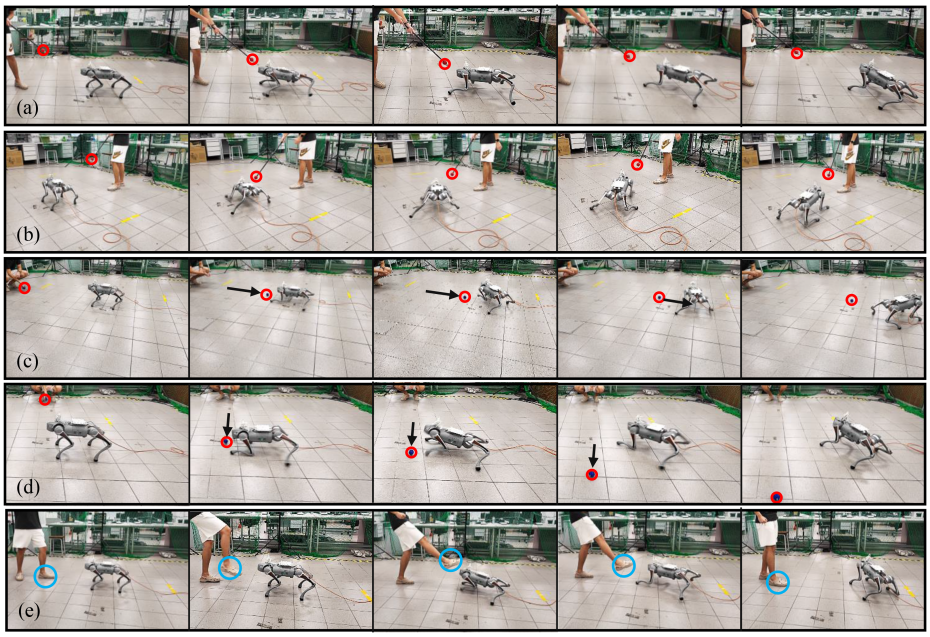


Fig. 1. The Reflexive Evasion Robot (**REBot**) system achieves instantaneous dynamic obstacle avoidance. When the fast-moving obstacles approach the quadrupedal robots (reaction time$<$ 1.5 s), REBot switches to the avoidance policy and performs reflexive evasion maneuvers. In (a) and (b), the robot is poked on the frontal and dorsal sides using a stick; in (c) and (d), a ball is launched toward the robot from both frontal and lateral directions; in (e), to evaluate robustness, the quadrupedal robot was subjected to intentional kicks from multiple directions. Experimental results demonstrate that the REBot system successfully controlled the quadrupedal robot to avoid all obstacles.

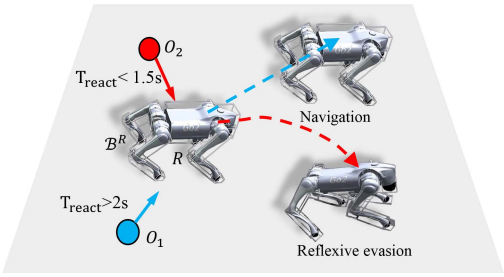


Fig. 2. Robot dynamic obstacle avoidance based on different reaction time. Red line: reflexive evasion under shorter reaction time; blue line: navigation_based avoidance under longer reaction time.

- We design the REBot system as a finite-state machine integrating avoidance and recovery policies to achieve robust, real-time reflexive evasion.
- We conduct comprehensive simulations and real-world experiments with thorough analysis to validate REBot's effectiveness across various obstacle scenarios.

## II. Related Works

Quadrupedal robots have achieved significant progress in recent years [17], [18], enabled by both model-based and learning-based control frameworks. Model-based approaches such as zero-moment point (ZMP) planning [19], [20], centroidal dynamics optimization [21], [22], and Model Predictive Control (MPC) [23], [24] enable precise trajectory tracking and support dynamic maneuvers under unstructured environments. In parallel, RL has empowered quadrupeds with a wide range of locomotion skills [14], from diverse gaits [25] to robust terrain traversal [26] and morphology adaptation [27]. RL has also been applied to static obstacle avoidance [8], where methods like ABS combine learned control with navigation planning to produce collision-free paths. However, such methods rely heavily on trajectory replanning and sufficient reaction time, which limits their applicability under unpredictable conditions.

Dynamic obstacle avoidance has been extensively studied across UAVs [3], [9], [28], [29], mobile robots [1], [30], humanoids [2], [31], and manipulators [32], [33]. UAVs leverage high-frequency planning and reactive control [3], [9], while Fan et al. [29] proposed a deep RL-based policy that enables fast, map-free evasive maneuvers against dynamic threats, even at high relative speeds. In mobile platforms, RL frameworks with motion constraints [1] improve navigation in crowded environments. For humanoids and manipulators, full-body coordination has been exploited to intercept or avoid moving objects [2], [33]. These advances demonstrate diverse platform-specific strategies for dynamic avoidance, but few translate directly to legged locomotion.

Despite these advances, quadrupedal robots currently lack general-purpose frameworks for dynamic obstacle avoidance under constrained reaction time. Existing approaches often rely on predefined motion primitives such as sidestepping, without the reflexive strategies seen in other platforms. Bridging this gap remains an open challenge and is critical for deploying legged robots in fast-changing environments.

## III. Preliminary

*Problem formulation:* Fig. 2 shows that the dynamic obstacle avoidance (DOA) system has two entities: dynamic obstacles

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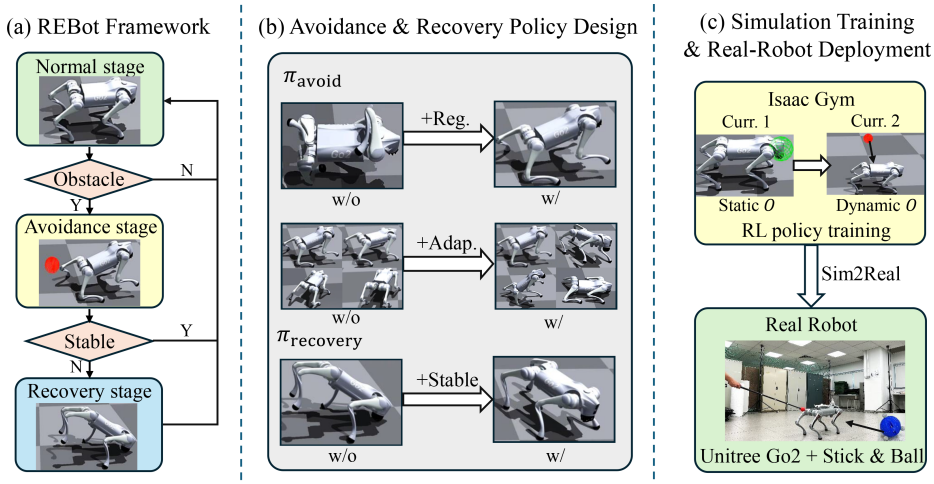


Fig. 3. **(a) REBot Framework:** A finite-state machine (FSM) governs transitions between the Normal, Avoidance, and Recovery stages. The quadrupedal robot performs reflexive evasion to avoid obstacles (red balls). **(b) Policy Design:** The avoidance policy is trained not only for successful obstacle avoidance but also incorporates regulation rewards for state stabilization and adaptive rewards to encourage diverse evasive behaviors. **(c) Training and Deployment:** REBot is trained in Isaac Gym using a two-stage curriculum from static to dynamic obstacles, and deployed on a real Unitree Go2 robot.

and quadruped robots. The dynamic obstacles $O$ are modeled as a rigid sphere with states $(r^O, p_t^O, v_t^O)$ of radius, position, velocity and acceleration in 3D space. The quadruped robot $R$ is a high-dimensional articulated system with a robot base and four independently actuated legs. The states $s_t^R$ consist of base position $p_t^R \in \mathbb{R}^3$, linear velocity $v_t^R \in \mathbb{R}^3$, angular velocity $\omega_t^R \in \mathbb{R}^3$, joint position $q_t^R \in \mathbb{R}^{12}$, joint velocity $\dot{q}_t^R \in \mathbb{R}^{12}$, joint torque $\tau_t^R \in \mathbb{R}^{12}$, contact force $f_t^R \in \mathbb{R}^{4\times3}$ and orientation angle $\theta_t^R \in \mathbb{R}^3$. The robot is driven by servo motors on the joints to move on the ground via action $a_t^R$, where $a_t^R \in \mathbb{R}^{12}$ denotes the joint target angles. In this work, we utilize the Unitree Go2 robot to conduct experiments. We define a successful dynamic obstacle avoidance (DOA) as the robot maintaining collision-free motion throughout the task duration. A collision is considered to occur if the signed distance function (SDF) from the obstacle center $p_t^O$ to the robot's oriented bounding box (OBB), denoted as $\mathcal{B}^R$, is smaller than the obstacle's radius $r^O$; that is, if $d(p_t^O, \mathcal{B}^R) < r^O$.

*Dynamic obstacle Avoidance Strategies:* The robot adopts a control system with observation $(p_t^R, \omega_t^R, q_t^R, \dot{q}_t^R, \tau_t^R, f_t^R, p_t^O, v_t^O, r^O)$ and action $(a_t^R)$, enabling the robot to avoid approaching obstacles while maintaining balance. When faced with static or slow-moving obstacles, the robot can temporarily stop and replan its trajectory, a behavior classified as **navigation avoidance** [8]. However, when the obstacles approach instantaneously, the robot must react immediately. We categorize such behavior as **reflexive evasion** (Fig. 2). The reaction time $T_{\text{react}}$ determines the reaction of navigation or reflex, and we distinguish these two behaviors by the maximum joint power of the robot. In the following sections, we design the REBot system to achieve the reflexive DOA and analyze the behaviors in both simulation and real-world experiments.

## IV. Method

In this section, we design the reflexive evasion robot (**REBot**) system to achieve DOA in Fig. 3(a). As shown in Fig. 3, the REBot system consists of three behavioral stages organized as a finite state machine (FSM). In the following sections, we introduce the FSM stages and transition criteria (Section IV-A), the training strategies of the avoidance policy (Section IV-B) and the recovery policy (Section IV-C). Finally, we delineate the training and deployment of the REBot system in simulation and real robots, respectively (Section IV-D).

### A. REBot Stages and Transition Criteria

The robot initially stays in the **normal stage** with functional behaviors such as standing, walking, or trotting. During this stage, the robot continuously observes the environment and potential obstacles. In this work, we define the normal behavior as standing still while maintaining balance via the PD controller. When the obstacle is approaching the robot, (i.e., $\langle v_t^O,\, p_t^R - p_t^O \rangle > 0$), REBot switches to the **avoidance stage** to execute reflexive evasion, where the avoidance policy performs reactive maneuvers under limited reaction time.

However, severe evasion may cause the robots' instability. Therefore, the REBot correspondingly switches to the **recovery stage**, where a recovery policy drives the robot back to normal functions. REBot judges the instability with three criteria: (i) body orientation exceeds a safe range $\|\theta_t^R\| > \theta_{\text{th}}^R$; (ii) joint velocity surpasses a stability limit $\|\dot{q}_t^R\| > \dot{q}_{\text{th}}^R$; (iii) base height drops below a threshold $h_t^R < h_{\text{th}}^R$. Here, $\theta_t^R$, $\dot{q}_t^R$, and $h_t^R \in \mathbb{R}$ denote the robot's orientation, joint velocity, and base height, respectively. The corresponding thresholds are $\theta_{\text{th}}^R$, $\dot{q}_{\text{th}}^R$, and $h_{\text{th}}^R$.

### B. Avoidance Policy

The avoidance policy is trained via RL to achieve reflexive evasion under constrained reaction time. The objective is to avoid collisions while maintaining postural stability and minimizing energy consumption. The reward function consists of three parts: $r = r_{\text{avoidance}} + r_{\text{regularization}} + r_{\text{adaptive}}$.

**Avoidance reward** is composed of $r_{\text{avoidance}} = r_{\text{distance}} + r_{\text{collision}}$. The distance reward is defined as $r_{\text{distance}} =$

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TABLE I
SUMMARY OF AUXILIARY REGULARIZATION TERMS

| Term | Purpose |
|---|---|
| $\lvert v_t^{R,z}\rvert^2$ | Penalize vertical velocity |
| $\lVert \dot{\theta}_t^{R,xy}\rVert$ | Penalize horizontal angular velocity |
| $\lVert \theta_t^{R,xy}\rVert$ | Penalize non-flat orientation |
| $\sum_i (a_t^{R,i} - a_{t-1}^{R,i})^2$ | Penalize abrupt action changes |
| $\sum_i \mathbf{1}_c^i$ | Penalize body collisions |
| $\lVert v_t^{R,xy} - v_t^{R,xy,\mathrm{cmd}}\rVert$ | Track command linear velocity |
| $\lvert \omega_t^{R,z} - \omega_t^{R,z,\mathrm{cmd}}\rvert$ | Track command angular velocity |
| $\sum_i (t_{\mathrm{air}}^i - 0.5)$ | Reward long foot swing phases |
| $\mathbf{1}\left(\max_i \left(\lVert \mathbf{f}_t^{R,xy,i}\rVert / \lvert f_t^{R,z,i}\rvert\right) > 5\right)$ | Penalize stumbling events |
| $\sum_i \lVert f_t^{R,i} - f_{\mathrm{th}}^{R,i}\rVert^2$ | Penalize excessive foot contact forces |

TABLE II
SUMMARY OF PPO TRAINING HYPERPARAMETERS

| Hyperparameter | Value |
|---|---|
| Actor hidden layers | [512, 256, 128] |
| Critic hidden layers | [512, 256, 128] |
| Activation | ELU |
| Learning rate ($\alpha$) | $1 \times 10^{-3}$ |
| Clip parameter ($\epsilon$) | 0.2 |
| Value loss coefficient ($c_1$) | 1.0 |
| Entropy coefficient ($c_2$) | 0.01 |
| Discount factor ($\gamma$) | 0.99 |
| GAE parameter ($\lambda$) | 0.95 |
| Desired KL divergence | 0.01 |
| Max gradient norm | 1.0 |
| Steps per env per iter | 24 |
| Mini-batches per iter | 4 |
| Learning epochs per iter | 5 |
| Max iterations | 5000 |
| Parallel environments | 4096 |

$-\exp(-(d(p_t^O, \mathcal{B}^R) - r^O))$ to encourage the robot to maintain a safe distance from the moving obstacle throughout the entire task. The collision penalty is defined as $r_{\mathrm{collision}} = \mathbf{1}(c=0) - \mathbf{1}(c=1)$ to penalize any contact event, where $c \in \{0, 1\}$ denotes the collision. We adopt a two-stage curriculum to improve policy training efficiency. In the first stage, a static obstacle appears instantaneously in a random location near the robot at a predefined activation time. In the second stage, the obstacle follows a directed trajectory toward the robot at varying speeds, simulating realistic threats.

**Regularization reward** consists of three terms $r_{\mathrm{regularization}} = r_{\mathrm{walk}} + r_{\mathrm{energy}} + r_{\mathrm{contact}}$. It is designed to ensure the learned evasive maneuvers remain both stable and natural (Fig. 3(b)). To promote natural and coordinated motion, we encourage symmetric limb phasing consistent with a trot gait. The term $r_{\mathrm{walk}} = \frac{1}{2}(\mathbf{1}(c_{\mathrm{FL}} = c_{\mathrm{RR}}) + \mathbf{1}(c_{\mathrm{FR}} = c_{\mathrm{RL}}))$ rewards synchronized contact patterns between diagonal leg pairs, where $c_{i,j}$ denotes different contact leg. We penalize the product of joint torque and joint velocity across all actuated degrees of freedom to reduce excessive power consumption through $r_{\mathrm{energy}} = -\sum_i |\tau_t^{R,i} \cdot \dot{q}_t^{R,i}|$, where $\tau_t^{R,i}$ and $\dot{q}_t^{R,i}$ represent the torque and angular velocity of each joint respectively. We also penalize the temporal fluctuation of vertical foot contact forces to reduce instability via $r_{\mathrm{contact}} = -\sum_i (f_t^{R,i,z} - f_{t-1}^{R,i,z})^2$, where $f_t^{R,i,z}$ denotes vertical foot contact force of each leg. These auxiliary regularization terms are summarized in Table I.

TABLE III
DOMAIN RANDOMIZATION FOR POLICY TRAINING

| Term | Value |
|---|---|
| **Observation** | |
| Joint position noise | $\mathcal{U}(-0.01, 0.01)$ rad |
| Joint velocity noise | $\mathcal{U}(-1.5, 1.5)$ rad/s |
| Angular velocity noise | $\mathcal{U}(-0.2, 0.2)$ rad/s |
| Projected gravity noise | $\mathcal{U}(-0.05, 0.05)$ m/s$^2$ |
| Height measure noise | $\mathcal{U}(-0.1, 0.1)$ m |
| **Dynamics** | |
| Friction factor | $\mathcal{U}(0.5, 1.25)$ |
| Added base mass | $\mathcal{U}(-1.0, 1.0)$ kg |
| Obstacle position (per axis) | $\mathcal{U}(-0.4, 0.4)$ m |
| Obstacle radius | $\mathcal{U}(0.05, 0.3)$ m |
| Obstacle velocity | $\mathcal{U}(1.0, 6.0)$ m/s |
| Reaction time | $\mathcal{U}(0.1, 4.0)$ s |
| **Episode** | |
| Episode length | $\mathcal{U}(8.0, 10.0)$ s |
| Command robot yaw | $\mathcal{U}(-1.0, 1.0)$ rad |
| Command robot velocity | $\mathcal{U}(-1.0, 1.0)$ m/s |
| Command robot heading | $\mathcal{U}(-\pi, \pi)$ rad |

**Adaptive reward** is designed as $r_{\mathrm{adaptive}} = r_{\mathrm{diversity}} + r_{\mathrm{threat}} + r_{\mathrm{direction}}$ to encourage motion diversity, speed modulation, and direction efficiency [34]. RL policy tends to converge toward a single locally optimal behavior (Fig. 3(b)). In this task, it manifests as the robot repeatedly using a fixed evasion gait regardless of obstacle state. We defined a diversity reward to encourage the policy to appropriate behaviors $r_{\mathrm{diversity}} = \mathrm{Var}_{s^R \sim \mathcal{D}_{s^R}}[\pi(a_t^R | s_t^R)]$. We define the threat level of the obstacle through the reaction time and the robot learns to adapt its speed in response to the levels of perceived threat as $r_{\mathrm{threat}} = -\lVert v_t^{\mathrm{R}} - v_t^{\mathrm{R,safe}}\rVert$, $v_t^{\mathrm{R,safe}} = v_t^{\mathrm{R,cmd}} + \lambda \exp(-\eta T_{\mathrm{reaction}})$, where $v_t^{\mathrm{R,cmd}}$ denotes the command velocity, $\lambda$ and $\eta$ denote the hyperparameters. To discourage evasive movements that deviate unnecessarily from the ideal escape direction, we penalize wrong movement direction via $r_{\mathrm{direction}} = -\langle v_t^R,\, p_t^O - p_t^R \rangle$.

### C. Recovery Policy

The recovery policy ensures a smooth transition from the avoidance stage back to the normal stage, allowing the robot to regain balance (Fig. 3(b)). Therefore, the reward function $r = r_{\mathrm{orientation}} + r_{\mathrm{stable}} + r_{\mathrm{position}} + r_{\mathrm{additional}}$ is designed corresponding to the instability criteria. The orientation reward defined as $r_{\mathrm{orientation}} = -\sum_i (\theta_t^{R,i} - \theta_0^{R,i})^2$ penalizes excessive tilt of the robot, where $\theta_0^{R,i} \in \mathbb{R}$ denotes the default orientation angles. The stable reward $r_{\mathrm{stable}} = \sum_i \exp(-|\dot{q}_t^{R,i}|)$ encourages low joint velocities via exponential decay. And the position term $r_{\mathrm{position}} = -\lVert p_t^R - p_0^R\rVert^2$ penalizes too slow base height, where $p_0^R$ denotes the default position. The additional reward term $r_{\mathrm{additional}}$ includes penalties on large joint torque and action discontinuities. These components are designed to reduce abrupt joint movements and encourage smoother transitions.

### D. Training in Simulation and Real-Robot Deployment

We implement the REBot system on the Unitree Go2 quadrupedal robot (Fig. 3(c)). We train the avoidance and recovery policies in the Isaac gym simulator [35] with PPO algorithm [36]. The hyperparameters shows in Table II. In addition,

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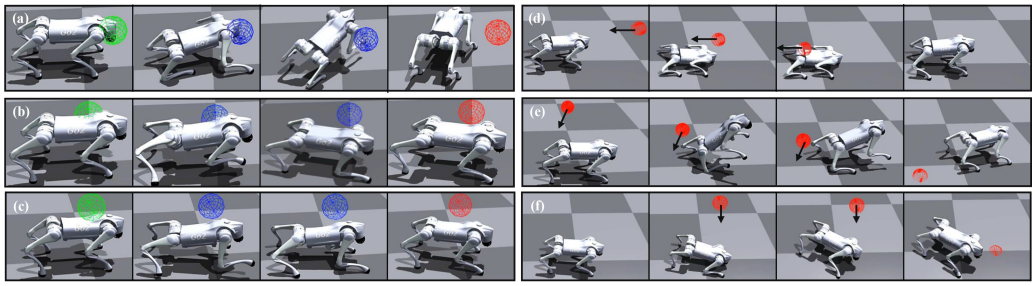


Fig. 4. Illustration of simulation experiments. **Curriculum 1**: Static obstacle appears with a delayed time. ● robot in normal stage; ● robot avoiding the obstacle; ● the obstacle appears and the robot evades the forbidden region. **Curriculum 2**: The robot avoids the red fast-moving obstacle at all times.

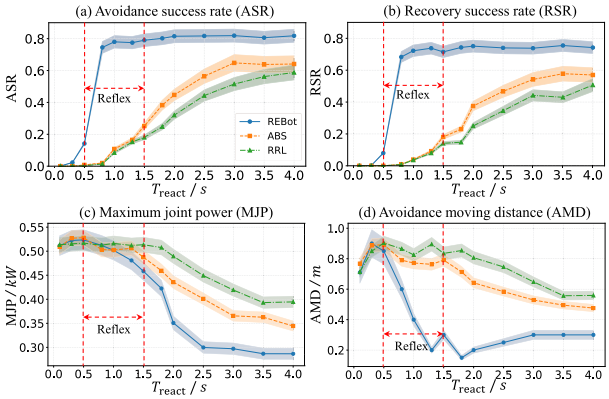


Fig. 5. Performances over Reaction Time. Red dashed region indicates reflexive evasion; longer reaction time corresponds to navigation-based avoidance.

we apply domain randomization: key observation, dynamics, and episode parameters are sampled uniformly at the start of each episode to improve robustness and transfer (Table III). Specifically, the avoidance policy is trained in two curricula. First, a stationary obstacle is randomly placed around the robot and activates after a delay; Go2 must react within the available response time. Second, a moving obstacle approaches with fixed velocity from a random direction, requiring real-time evasion. Both curricula randomize obstacle parameters to prevent overfitting and encourage generalization across planning-based and reflexive behaviors. Then we deploy the REBot system to the real Unitree Go2 robot. A motion capture system was used to provide real-time position data for both the robot and the dynamic obstacle. To emulate dynamic obstacles, we used a rigid rod with a lightweight ball attached to its tip, serving as a physical proxy for an incoming object. The obstacle's position was continuously tracked via reflective markers. The real-robot experiment shares the same evaluation metrics as simulation. For each test, we ran 50 randomized trails and report the results as mean with 95% confidence intervals.

## V. Simulation Experiments

We validate and estimate the performance of REBot in the simulation system. In this section, we answer three questions: **Q1** Can REBot achieve successful evasion under instantaneous DOA? (Section V-B) **Q2** What are the robots' reactions under different obstacle conditions? (Section V-C) **Q3** How can the rewards' design and recovery stage influence DOA performance? (Section V-D)

TABLE IV
Simulation Experiment Results

| $T_{\text{react}}$ / s | Metric | ABS$^{\diamond}$ | RRL$^{\diamond}$ | REBot |
|---|---|---|---|---|
| 0.1 ∼ 0.5 | ASR↑$^{*}$ | 0.00 | 0.00 | **0.05** |
| | RSR↑$^{*}$ | 0.00 | 0.00 | **0.03** |
| | MJP↓$^{*}$ | 0.51 | 0.52 | **0.50** |
| | AMD↓$^{*}$ | 0.84 | 0.85 | **0.82** |
| 0.5 ∼ 1.5 | ASR↑ | 0.11 | 0.09 | **0.65** |
| | RSR↑ | 0.06 | 0.05 | **0.59** |
| | MJP↓ | 0.52 | 0.51 | **0.49** |
| | AMD↓ | 0.80 | 0.86 | **0.47** |
| 1.5 ∼ 4.0 | ASR↑ | 0.51 | 0.41 | **0.81** |
| | RSR↑ | 0.42 | 0.32 | **0.74** |
| | MJP↓ | 0.40 | 0.45 | **0.34** |
| | AMD↓ | 0.60 | 0.70 | **0.26** |

$^{*}$ ASR: avoidance success rate; RSR: recovery success rate; MJP: maximum joint power; AMD: avoidance moving distance.
$^{\diamond}$ ABS: Agile But Safe [8]; RRL: Reactive RL [29].

### A. Experiment Settings

*Tasks:* We conducted simulation experiments in the Isaac gym [35] to evaluate the DOA ability of the REBot system. During testing, the obstacle approaches from diverse directions within a 180° arc in the XZ, YZ, and XY planes of the robot's body frame (Fig. 6), covering frontal, lateral, overhead, and ground-level threats. The response time is expanded beyond training, with $T_{\text{react}} \in [0.1, 4.0]$ s, allowing evaluation across both immediate reaction and delayed planning scenarios.

*Metrics:* The systems are evaluated with five metrics. The avoidance success rate (*ASR*): $N_{\text{avoid}}/N_{\text{total}}$; the recovery stability rate (*RSR*): $N_{\text{recover}}/N_{\text{avoid}}$, indicating the proportion of trials where the robot successfully stabilizes after avoidance; maximum joint power (*MJP*); avoidance moving distance (*AMD*): the base displacement between the robot's initial and final positions; and gait diversity index (*GDI*): $\mathbb{E}_{s^R\sim\mathcal{D}(s^R)}[\text{Var}_{a^R\sim\pi(a^R|s^R)}[a^R]]$ the expected action variance under the learned policy, where

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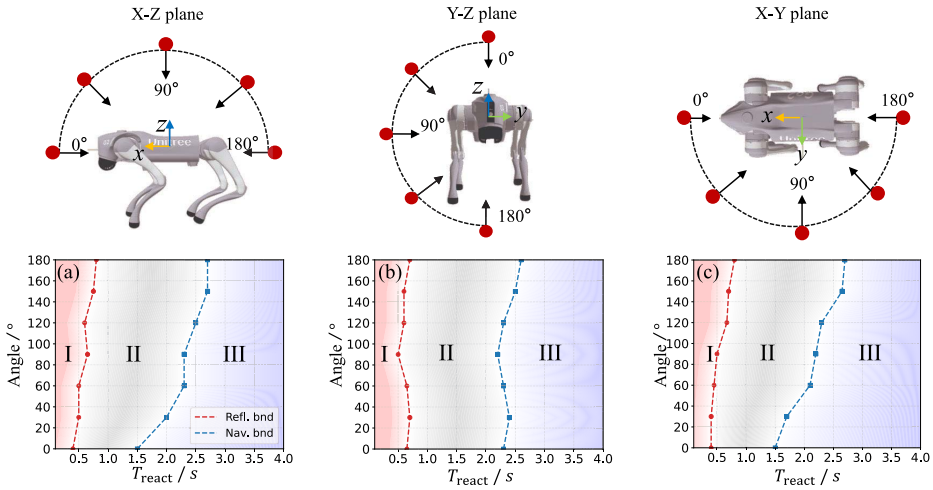


Fig. 6. Robot avoids obstacles in various directions and reaction time. **Top row** shows the obstacles' approaching directions in three planes. **Bottom row** figures indicate the different avoidance behaviors. Region I: avoidance failure, II: reflexive evasion, III: navigation avoidance.

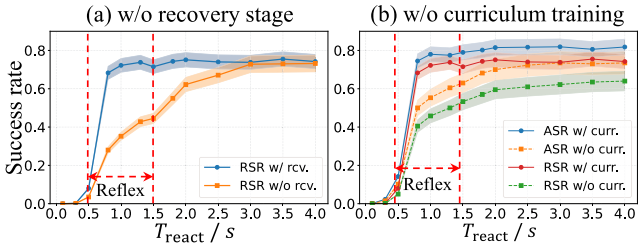


Fig. 7. Success Rate of Ablation Studies. Red dashed region indicates reflexive evasion; longer reaction time corresponds to navigation-based avoidance.

$\pi(a^R|s^R)$ denotes the policy distribution over actions at state $s^R$, and $\mathcal{D}(s^R)$ is the state distribution collected during execution.

*Baselines:* 1) Agile But Safe (ABS) [8] achieved robust static obstacle avoidance with high-speed navigation motions, without the capability for dynamic obstacles. 2) Reactive RL (RRL) [29] is developed for dynamic obstacle avoidance in the UAV system. The avoidance strategy is based on simplified rigid-body dynamics, which do not generalize to legged whole-body systems.

### B. Main Experimental Results

The REBot system is trained in static obstacle and dynamic obstacle avoidance curricula. Fig. 4 visualizes the simulation experimental results with different obstacle conditions and avoidance behaviors. When obstacles appear in front or on the sides, the robot tends to jump away for evasion (Fig. 4(a), (b), (e)). When obstacles appear on top, the robot will crouch down (Fig. 4(c), (d), (f)). These results demonstrate that REBot enables the robot to select appropriate avoidance strategies based on the obstacle state.

We categorize the reaction time into three intervals (Table IV and Fig. 5(a), (b)): 0.1 $\sim$ 0.5 s, 0.5 $\sim$ 1.5 s and 1.5 $\sim$ 4.0 s. In the first interval, all methods obtain nearly 0 success rates due to insufficient reaction time. In the second interval with moderately short reaction times, REBot shows reflexive evasions, while both baselines still take low-speed navigation gaits. This difference yields much higher ASR and RSR of REBot. In the third interval, REBot remains high ASR and RSR. ABS and RRL also improve their performance due to the longer reaction time, but still fall short of REBot because they are not specialized for active DOA. The overall trend, as illustrated in Fig. 5, indicates that REBot effectively addresses instantaneous DOA challenges.

We also explore the relationship between MJP, AMD and reaction time (Fig. 5(c), (d)). When the reaction time is extremely short (e.g., below 0.5 s), we observe a high MJP over 500 W and a large AMD, caused by intense evasion behaviors such as jumping away. In the moderately short reaction time interval, both MJP and AMD decrease, where REBot can adopt more appropriate avoidance behaviors such as crouching down. With longer reaction time, both MJP and AMD converge to lower values, as REBot has enough time to execute smoother navigation-based avoidance behaviors. The trends show that REBot adapts avoidance behaviors based on time constraints and avoidance efficiency.

### C. Analysis of Avoidance Ability

We evaluate the effect of the obstacle direction on the robot's avoidance ability by applying impacts from different angles within the X-Z, Y-Z and X-Y planes of the robot's body frame (Fig. 6). Based on ASR and MJD, we divide the robot's avoidance behavior space into three regions: region I, where the robot fails to avoid; region II, where the robot adopts reflexive evasion; and region III, where the robot adopts navigation-based avoidance. The boundary between region I and II is defined by ASR over 30%, while the boundary between region II and III is defined by MJD below 300 W.

In the X-Z and X-Y planes (Fig. 6(a), (c)), front obstacles are easier to avoid, and navigation-based avoidance can be achieved with shorter reaction time. In contrast, obstacles approaching from the back require longer reaction time and make navigation-based avoidance more difficult. This asymmetry stems from mechanical design of Unitree Go2, where the leg structure favors faster backward motion but makes forward jumping harder. In the Y-Z plane (Fig. 6(b)), we find that obstacles approaching

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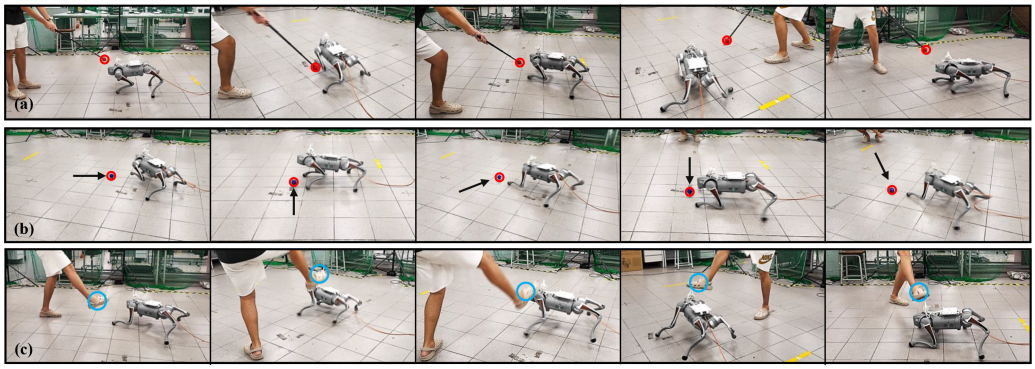


Fig. 8. REBot system real-robot demonstrations on Unitree Go2 Robot (See video). (a) the robot is poked from different directions using a stick; (b) a ball is launched towards the robot from different directions; (c) the robot is subjected to intentional kicks from different directions.

TABLE V
ABLATION PERFORMANCES OF REBOT

| $T_{\text{react}}$ / s | Metric | w/o rcv.[1] | w/o curr.[2] | w/o adp.[3] | REBot |
|---|---|---|---|---|---|
| 0.5∼1.5 | ASR↑* | 0.63 | 0.48 | 0.59 | **0.65** |
| | RSR↑* | 0.31 | 0.39 | 0.51 | **0.59** |
| | GDI↑* | 2.46 | 2.41 | 1.43 | **2.51** |
| 1.5∼4.0 | ASR | 0.80 | 0.71 | 0.78 | **0.81** |
| | RSR | 0.63 | 0.60 | 0.69 | **0.74** |
| | GDI | 2.06 | 2.24 | 1.36 | **2.13** |

* ASR: avoidance success rate; RSR: recovery success rate; GDI: gait diversity index.
[1] w/o recovery stage; [2] w/o curriculum one learning; [3] w/o adaptive reward in avoidance policy.

from the sides are easier to avoid compared to those from the top or bottom, with a shorter transition time from reflexive evasion to navigation-based avoidance. These results highlight the robot's avoidance ability varies significantly depending on the obstacle direction.

### D. Ablation Studies of REBot System

*The recovery stage ensures a stable standing posture after rapid reflexive evasion:* To validate its effectiveness, we compare the performance of the REBot system with and without the recovery policy. As shown in Table V and Fig. 7(a), removing the recovery stage leads to a drop (20%) in the success rate within the reflex region. Additionally, as reaction time increases, avoidance behaviors shift from reflex to navigation, reducing the influence of the recovery stage on robot stabilization.

*Curriculum learning enables a smooth transition from normal stage to fast-moving reflexive evasion:* In the first stage, the robot learns to avoid sudden obstacles at random positions; in the second stage, it generalizes to obstacles approaching from diverse directions. An ablation study reveals the importance of this progressive training: when policies are trained directly on the second curriculum (skipping the first), Table V and Fig. 7(b) show removing the first curriculum causes a 5% drop in ASR and RSR for both reflexive evasion and navigation avoidance. This gap stems from moderate and diverse gaits learned during static obstacle avoidance, enabling a stable start against fast-moving obstacles.

*The adaptive reward encourages diversified avoidance gaits that improve avoidance robustness:* Without this term, the reinforcement learning algorithm converges to a single avoidance behavior (e.g., consistently jumping backward) regardless of obstacle variations. We conduct an ablation study to evaluate the adaptive reward's effectiveness. As shown in Table V, removing the adaptive reward results in: (1) a significant 40% decrease in gait diversity index (GDI), demonstrating reduced behavioral diversity, and (2) moderate declines in both ASR and RSR. These findings indicate that the gait diversity promoted by the adaptive reward contributes directly to improved avoidance performance.

## VI. REAL-ROBOT DEMONSTRATION

We deploy the REBot system on a Unitree Go2 robot to demonstrate its reflexive evasion capabilities in real-world scenarios. The robot and obstacles are tracked using an OptiTrack motion capture system to provide accurate position information. To generate diverse dynamic obstacles, we test three interaction types: poking with a stick (Fig. 8(a)), throwing a ball (Fig. 8(b)), and kicking (Fig. 8(c)), each targeting the robot from different directions, including front, left, right, left-front, and right-front.

When an obstacle approaches, the robot enters avoidance mode to evade reflexively. The primary avoidance actions include jumping away and crouching down. After completing the avoidance maneuver, the robot enters recovery mode to regain a stable stance. Additionally, we observe that when the poking motion is relatively slow, the robot tends to adopt navigation-based avoidance over reflexive actions, leveraging longer available reaction time for smoother behaviors.Under the real-world test conditions, the REBot system achieves an ASR of 56% and an RSR of 53%. To contextualize the Sim2Real gap, we report quantitative execution factors from onboard signals: control response latency of 88 ms $\pm$ 7 ms (95% CI), a torque tracking gap (hardware vs. simulation) of $14.4\% \pm 2\%$ (95% CI), and an effective contact friction of $\mu_{\text{eff}} = 0.62 \pm 0.05$ (95% CI).

## VII. CONCLUSION

We initiate the study of reflexive evasion as a critical capability for dynamic obstacle avoidance in quadrupedal robots, where navigation-based methods fail under tight reaction constraints.

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To address this challenge, we develop REBot, a unified control system that couples rapid avoidance and stability recovery through RL and structured training. Extensive simulation and real-robot experiments show reliable, adaptive evasion, while revealing important characteristics of reflexive responses shaped by robot morphology and obstacle dynamics. Our results point to more agile, resilient, and safety-aware legged robotic for dynamic environments.

Despite its effectiveness, REBot has several limitations. First, REBot focuses on the locomotion-side reflexive evasion and assumes ground-truth obstacle states; replacing motion capture with independent onboard perception (e.g., LiDAR, camera) is planned for future work. Second, experiments are conducted in open space and the avoidance policy is unconstrained. Future work will integrate the planning layer to enforce environmental constraints by checking reachable space. Third, the avoidance behavior is biased by the hardware design, favoring backward jumps due to leg morphology. Lastly, a Sim2Real gap remains in low-level motor control, as joint velocity commands in simulation do not fully capture torque-servo dynamics in high-speed evasive maneuvers, nor the variability from real-world ground friction.

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